

Andrew Peterson

QUANTUM RESEARCHER · SECURITY ENGINEER · SOFTWARE ARCHITECT

California, United States of America

☎ (+1) 951-551-6946 | ✉ acpeters@ucsd.edu | 📱 andypeterson2 | 🌐 i-am-andy-peterson

Education

University of California: San Diego

BACHELORS OF SCIENCE IN COMPUTER SCIENCE

La Jolla, California

Sep. 2020 - Dec. 2024

Experience

Qualcomm Institute

QUANTUM SOFTWARE ENGINEERING LEAD (INTERN)

La Jolla, California

Jul. 2022 - Dec. 2024

- Architected and spearheaded a video communication platform utilizing end-to-end quantum encryption via photonic circuitry, enhancing data security protocols.
- Optimized Grover's algorithm using Qiskit to solve complex binary constraint satisfaction problems, demonstrating practical applications for quantum search.
- Developed and evaluated Quantum Machine Learning (QML) models, implementing classification algorithms via quantum neural networks to improve predictive accuracy.
- Mentored a cross-functional team of interns under Dr. Nikola Alic, overseeing software deliverables and the technical setup of a new undergraduate quantum computing lab.

Rochester Institute of Technology Esports

FULL-STACK DEVOPS ENGINEER & ADVISOR

Remote (Rochester, New York)

Sep. 2020 - Jul. 2022

- Engineered and deployed scalable web resources using the MERN stack (MongoDB, Express, React, Node.js), leveraging Docker for consistent environment orchestration.
- Orchestrated cloud infrastructure on DigitalOcean Droplets, implementing Nginx reverse proxies and Docker Compose to ensure 99.9% uptime for high-traffic services.
- Streamlined development lifecycles by building CI/CD pipelines and monitoring suites, reducing deployment friction and enhancing system performance.

Mathnasium

INSTRUCTOR & IT SYSTEMS SPECIALIST

Temecula, California

Mar. 2020 - Sep. 2021

- Delivered advanced technical instruction in Calculus and Computer Science, tailoring complex concepts to diverse student learning levels.
- Directed the center's digital transformation during COVID-19, authoring technical documentation and maintaining IT infrastructure to enable seamless remote learning.

Extracurricular

Quantum Computing at UC San Diego

CO-FOUNDER & PRESIDENT (2023 - 2024) & TECHNICAL CURRICULUM LEAD

La Jolla, CA

Feb. 2021 - Dec. 2024

- Co-founded and scaled the first undergraduate quantum computing organization at UC San Diego, establishing a community for 100+ aspiring researchers and engineers.
- Developed a comprehensive technical curriculum for workshops covering foundations in quantum computing and Qiskit.
- Standardized organizational workflows and digital infrastructure, enabling sustainable leadership transitions and cross-campus collaborations.
- Presented components of internship to organization members, including quantum-encrypted communication and quantum search algorithms.

ACM Cyber @ UCSD

PRESIDENT & CYBERSECURITY COMPETITION MEMBER

La Jolla, California

Aug 2020 - June 2023

- Trained teams for (and competed in) high-stakes national and international competitions including CCDC, CPTC, and Google CTF.
- Authored pedagogical training modules on penetration testing, network defense, and cryptography to elevate the club's competitive ranking.
- Orchestrated weekly technical workshops and "Deep Dive" sessions, fostering a culture of peer-to-peer mentorship in advanced security topics.

San Diego Capture the Flag (SDCTF)

LEAD INFRASTRUCTURE ENGINEER & CHALLENGE DESIGNER

Remote

Dec. 2020 - May 2022

- Engineered the backend for San Diego's largest annual cybersecurity competition, supporting 500+ global participants in its inaugural year.
- Provisioned cloud-native infrastructure using Terraform (IaC), managing a Postgres database and containerized challenge environments.
- Designed and deployed a Discord.JS-integrated submission bot to automate real-time flag validation and leaderboard updates.
- Developed high-quality Open Source Intelligence (OSINT) challenges, testing competitors' ability to perform investigative digital forensics.

Skills

Languages	Python (Expert), C/C++, Java, TypeScript/JavaScript, SQL, Bash, LaTeX, Matlab, Rust
Quantum Computing	Qiskit, Cirq, Quantum Neural Networks (QML), QAOA, Grover's & HHL Algorithms, BB84, Error Mitigation
Systems & DevOps	Docker, Kubernetes, Terraform, Linux Kernel, Distributed Systems, CI/CD, Nginx, Git, AWS, DigitalOcean
Cybersecurity	Reverse Engineering (Ghidra/IDA), Network Security (TCP/IP), Penetration Testing, OSINT, Applied Cryptography
Software Engineering	Data Structures & Algorithms (DSA), System Design, Design Patterns, Agile/Scrum, Unit Testing
Data Science & AI	PyTorch, Scikit-Learn, Pandas, Computer Vision, NLP, Ollama (LLMs), Statistical Modeling, Jupyter
Infrastructure	PostgreSQL, MongoDB, Redis, RESTful/GraphQL API Design, Microservices, Linux Server Administration
Experimental Lab	Free-space Photonics, Laser Systems, Signal Processing, Oscilloscopes, Atomic Force Microscopy (AFM)

Certifications

2025	Azure Data Fundamentals, Microsoft
2022	IT Fundamentals+, CompTIA

References

Dr. Nikola Alic	Internship Director at Qualcomm Institute	+1 (858) 228-7700	nalic@ucsd.edu
Ryan Gillie	Supervisor at RIT Esports	+1 (434) 251-6974	gillie.ryan@hotmail.com